



"Others detect the threat. NetraX investigates the threat."

NETRAX

AI-Powered Email Threat Detection, GeoLocation & Forensic Intelligence Platform

Detection tells you WHAT. NetraX investigates WHY.

PROBLEM STATEMENT ID	THEME	PS CATEGORY	TEAM ID	TEAM NAME
SIH26106	Blockchain & Cybersecurity	Software	[TEAM ID]	Nexora
PROBLEM STATEMENT TITLE				
AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform				

All India Council for Technical Education (Cyber Security Cell)

Decision-support & forensic-intelligence prototype — not a substitute for banks, police or official cybercrime reporting (cybercrime.gov.in) · SIH 2026



Detection tells you WHAT happened. NetraX investigates WHY — and hands you the evidence.

01 INGESTION

✉ SUSPICIOUS EMAIL (.eml / paste / JSON)

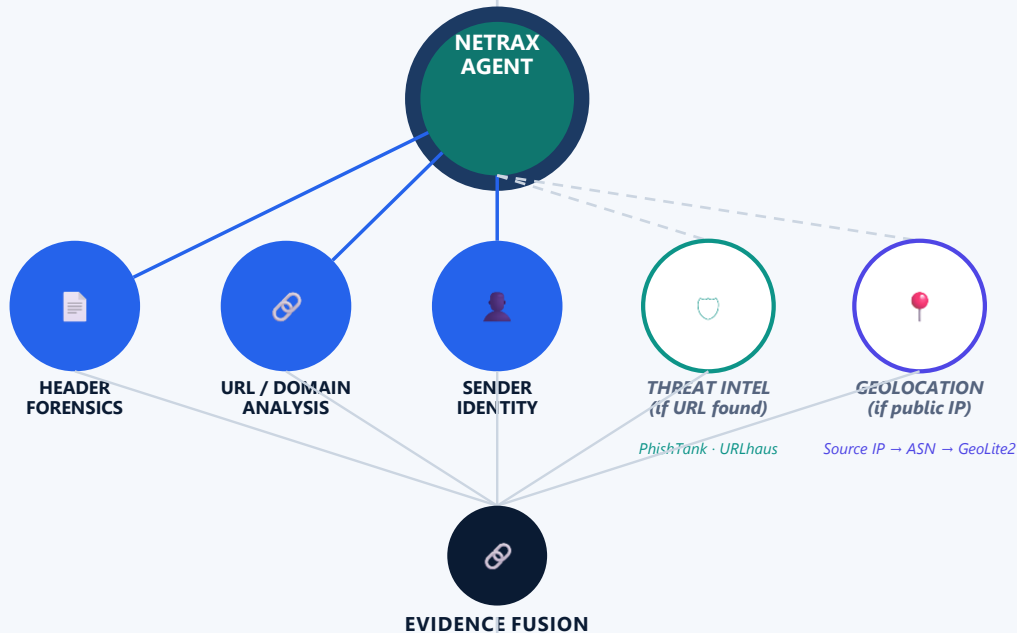
📧 EMAIL PARSER — MIME + indicator extraction

02 ORCHESTRATOR — decides which tools this email needs

03 TOOLS

● always run

○ conditional



RISK ENGINE
(0-100,
deterministic) →

LOW

MEDIUM

HIGH

CRITICAL

06 DECIDE

🗨 EXPLAINABLE FINDINGS

📄 CASE REPORT

FRONTEND

React · TypeScript · Vite · Tailwind CSS · shadcn/ui · Framer Motion · Recharts

BACKEND / ML

Supabase · PostgreSQL · Edge Functions · Python · scikit-learn · Gemini API (explanation layer, planned)

DATA GROUNDING

Enron Email Corpus — structural/language reference — not a labelled class

SpamAssassin Corpus — trains the email content classifier

UCI Phishing Websites — trains the URL classifier

PhishTank — phishing URL threat intel

URLhaus — malware-distribution URL threat intel

MaxMind GeoLite2 — IP → ASN → geolocation enrichment

TRAINED ML MODELS

Email content classifier

TF-IDF + Linear SVM · F1 = 0.961, ROC-AUC = 0.998
(measured, held-out SpamAssassin test set)

URL / domain classifier

HistGradientBoosting · F1 = 0.954
(measured, UCI Phishing Websites test set)

AGENTIC LOGIC

- No URL in email → URL & threat-intel tools skipped entirely
- No public source IP → geolocation skipped entirely
- Every skip is logged — auditable, not silent

AI orchestrates & explains. The risk engine calculates — never an LLM guessing a number.

01 PROTOTYPE (TODAY)

203 automated tests passing
(169 TS + 34 Python — measured)

02 VALIDATION

Threshold tuning · cross-validation
human review · feedback loop

03 DEPLOYMENT-READY

Portable Edge Functions · RLS schema
credential-gated live intel

CHALLENGE → MITIGATION

● Data quality

➡ ● Preprocessing + profiling before training (reports/data_quality_report.md)

● Threat-intel / API availability

➡ ● Adapter architecture; reports “unavailable”, never a fabricated match — cached URLhaus snapshot for demo

● ML false positives / negatives

➡ ● Cross-validated model selection, tuned decision threshold, held-out evaluation

● Unseen attack patterns

➡ ● Multi-signal fusion (headers + URL + ML + threat intel) + a documented feedback loop

● Computational requirements

➡ ● ML inference isolated in its own service; investigation logic is portable/modular

● Platform security

➡ ● Auth + row-level security · input validation · SSRF-safe URL fetches · no secrets in code

WHY IT IS FEASIBLE

✓ Public research datasets already acquired & profiled

✓ PhishTank / URLhaus integrated via tested adapters

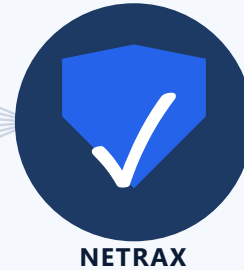
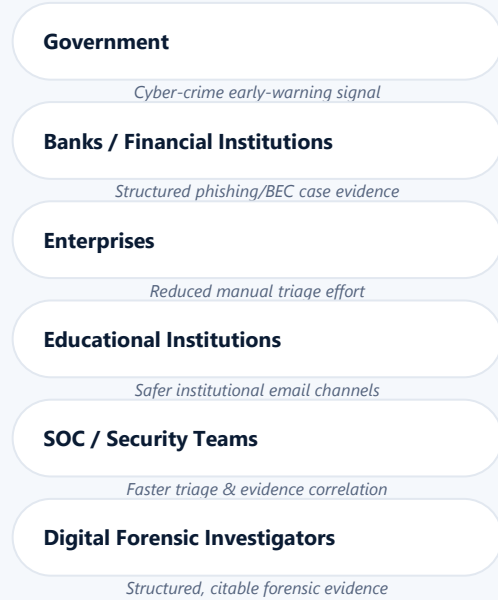
✓ GeoLite2 IP→ASN→geo enrichment adapter, unit tested

✓ ML inference kept modular — portable pipeline

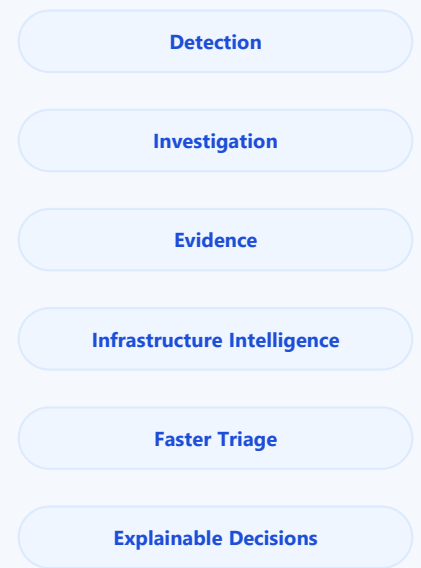
✓ Supabase/PostgreSQL schema scales case & evidence storage

✓ Live intel when credentialed; labelled demo data otherwise

WHO IT SERVES



WHAT CHANGES



A decision-support & forensic-intelligence platform — not a replacement for banks, police or official cybercrime systems.

BEFORE VS. AFTER NETRAX

BEFORE



AFTER



DATA & INTELLIGENCE

- **CMU Enron Email Dataset**
cs.cmu.edu/~enron
- **Apache SpamAssassin Public Corpus**
spamassassin.apache.org
- **UCI Phishing Websites Dataset**
archive.ics.uci.edu/dataset/327
- **PhishTank**
phishtank.org
- **URLhaus (abuse.ch)**
urlhaus.abuse.ch
- **MaxMind GeoLite2**
dev.maxmind.com/geoip/geolite2-free-geolocation-data

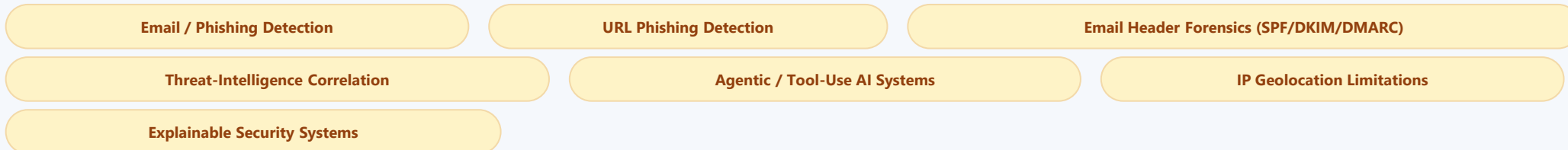
TECHNOLOGY

- **React + TypeScript + Vite**
react.dev
- **Tailwind CSS + shadcn/ui**
tailwindcss.com
- **Supabase / PostgreSQL**
supabase.com/docs
- **Python + scikit-learn**
scikit-learn.org
- **Gemini API**
ai.google.dev/gemini-api/docs
- **(explanation layer — planned, not yet wired)**

SECURITY & STANDARDS

- **OWASP Top 10 / SSRF Prevention**
owasp.org
- **NIST Cybersecurity Framework**
nist.gov/cyberframework
- **CERT-In Advisories**
cert-in.org.in
- **Govt. of India Cybercrime Portal**
cybercrime.gov.in

RESEARCH BASIS



Topic areas grounding this build — specific citations to be finalized for the final report.